Reg. No. | | | | | | | | | | |

[18MTADE202]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

WIND TUNNEL THEORY AND TESTING

Time: 03 Hours **Maximum Marks:100**

Instructions:

1. Answer **FIVE** full Questions.
2. Question numbers one and two are compulsory.
3. Missing data may be suitably assumed.

1.	a.	Explain the Buckingham pi theorem. Apply it to the case of uniform fluid flow around a smooth sphere.	10
	b.	Explain the different types of corrections required for 2-D flow in wind tunnel closed test sections.	10
2.	a.	Explain the purpose and features of water tunnels and their advantages as compared to wind tunnels.	10
	b.	Explain the purpose and features of -i) Icing Tunnel & ii) Gust Response tunnel	10
3.	a.	Explain the closed circuit subsonic wind tunnel components and their function. Draw a neat sketch.	10
	b.	Explain the features of settling chamber, contraction cone and diffuser in closed circuit subsonic wind tunnel.	10

OR

4	a.	Explain the need for a CD nozzle for supersonic flow in wind tunnels. Explain the different 'critical pressure ratios' in the functioning of a CD nozzle.	10
	b.	Explain the process of starting a supersonic wind tunnel and 'swallowing of the starting shock ' by the diffuser throat.	10
5.	a.	Explain external platform type wind tunnel force balance, its working and typical scheme for measurement of the six forces and moments.	10
	b.	Explain the working principle of any three types of advance force sensors used in wind tunnel balances	10

OR

6.	a.	Explain how flow direction is measured in wind tunnels	10
	b.	What are the different types of wind tunnel temperature sensors? Explain the working of Pyrometer.	10
7.	a.	Explain flow visualization and Stream lines, Streak lines, Path lines and Time lines.	10
	b.	Explain any four methods for surface flow visualization	10

OR

8.	a.	What do you understand by 'Tracer methods for flow visualization'? Explain any two techniques of tracer methods for flow visualization	10
	b.	What are the different types of optical methods for flow visualization? Explain the shadowgraph method.	10

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